

TITLE PAGE

- **Problem Statement ID:** 4
- **Problem Statement Title:** AI/ML-Based Behavioral Anomaly Detection for Cybersecurity
- **Theme:** Smart Automation & Cybersecurity
- **PS Category:** Software
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- **IDEA TITLE:** AI-Powered Behavioral Anomaly Detection for Cybersecurity

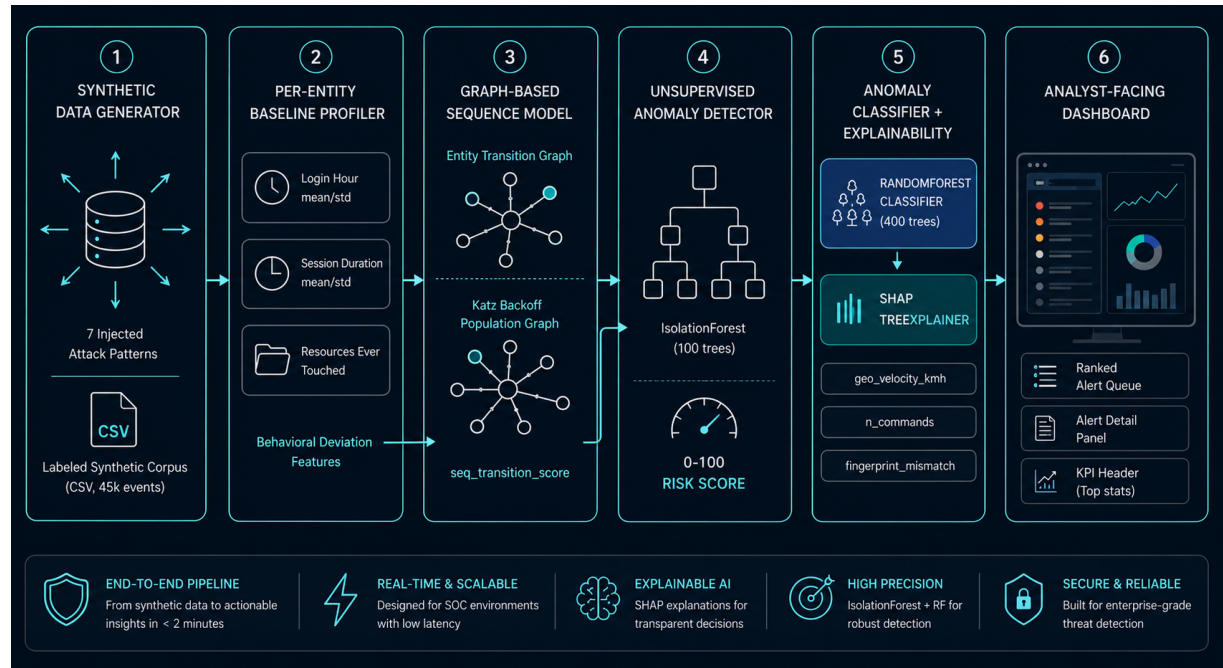
PROPOSED SOLUTION

- **Proposed Solution:** A near real-time AI system that learns "normal" behavioral fingerprints (login timing, location, resource usage) to detect cyber threats across IT, OT, and IoT edge devices.
- **Detailed explanation:** The system operates on a 6-stage pipeline. It profiles baseline behavior and builds an entity-resource sequence graph. This feeds 15 deviation features into an unsupervised IsolationForest to generate a 0-100 risk score, followed by a RandomForest classifier and SHAP to clearly explain the alert to analysts.
- **How it addresses the problem:** Traditional signature-based tools miss novel threats. By learning "normal" behavior per entity, this system catches 7 categories of advanced attacks (e.g., lateral movement, insider drift, low-and-slow exfiltration) without relying on known signatures.
- **Innovation and uniqueness:**
 - **Order-Aware:** Uses a graph transition model to score the *sequence* of actions, not just isolated snapshots.
 - **Explainable (XAI):** Provides human-readable, SHAP-explained reasons for every alert, eliminating "black box" scores.
 - **Resilient:** Explicitly handles "cold-starts" for new users and adapts to concept drift over time.

TECHNICAL APPROACH

Technologies used:

- Python
- NumPy
- pandas
- scikit-learn (IsolationForest, RandomForest)
- SHAP
- Faker
- HTML/JS analyst dashboard
- Node.js (docx/pptxgenjs) for reporting — no GPU or heavy dependencies required.



Methodology and process for implementation:

- *1. Synthetic Generator*: Creates access logs with hidden ground truth for testing.
- *2. Baseline Profiler*: Computes behavioral deviations from running history to prevent data leakage.
- *3. Sequence Graph*: Scores whether the *order* of user actions is anomalous.
- *4. IsolationForest (Detection)*: Evaluates 15 features across 300 trees to output a 0-100 risk score.
- *5. RandomForest & SHAP (Classify & Explain)*: Predicts the attack category and attributes it to specific factors for the analyst dashboard.
- *Design Tradeoff*: We deliberately chose a graph-based transition model over deep sequence encoders (like LSTMs) to prevent overfitting on the 45k event corpus.

FEASIBILITY AND VIABILITY

- **Analysis of feasibility**

Proven Accuracy: The working prototype achieves a 0.943 ROC-AUC, 76.6% recall, and 98.4% classifier accuracy.

High Efficiency: The lightweight stack trains on 45,580 events in minutes on a standard laptop, ensuring a highly scalable and cheap deployment.

- **Potential challenges and risks**

Data Limitations: Rule-based synthetic attacks are easier for models to learn than real-world adversarial behavior.

Ambiguous Threats: Edge cases like "insider drift" and "impossible travel" (sparse data) are inherently difficult to classify.

Stream Processing: The current model computes features in batch passes; concept drift is only partially handled.

- **Strategies for overcoming these challenges**

Pilot with real, anonymized SOC logs to refine models against true adversarial behavior.

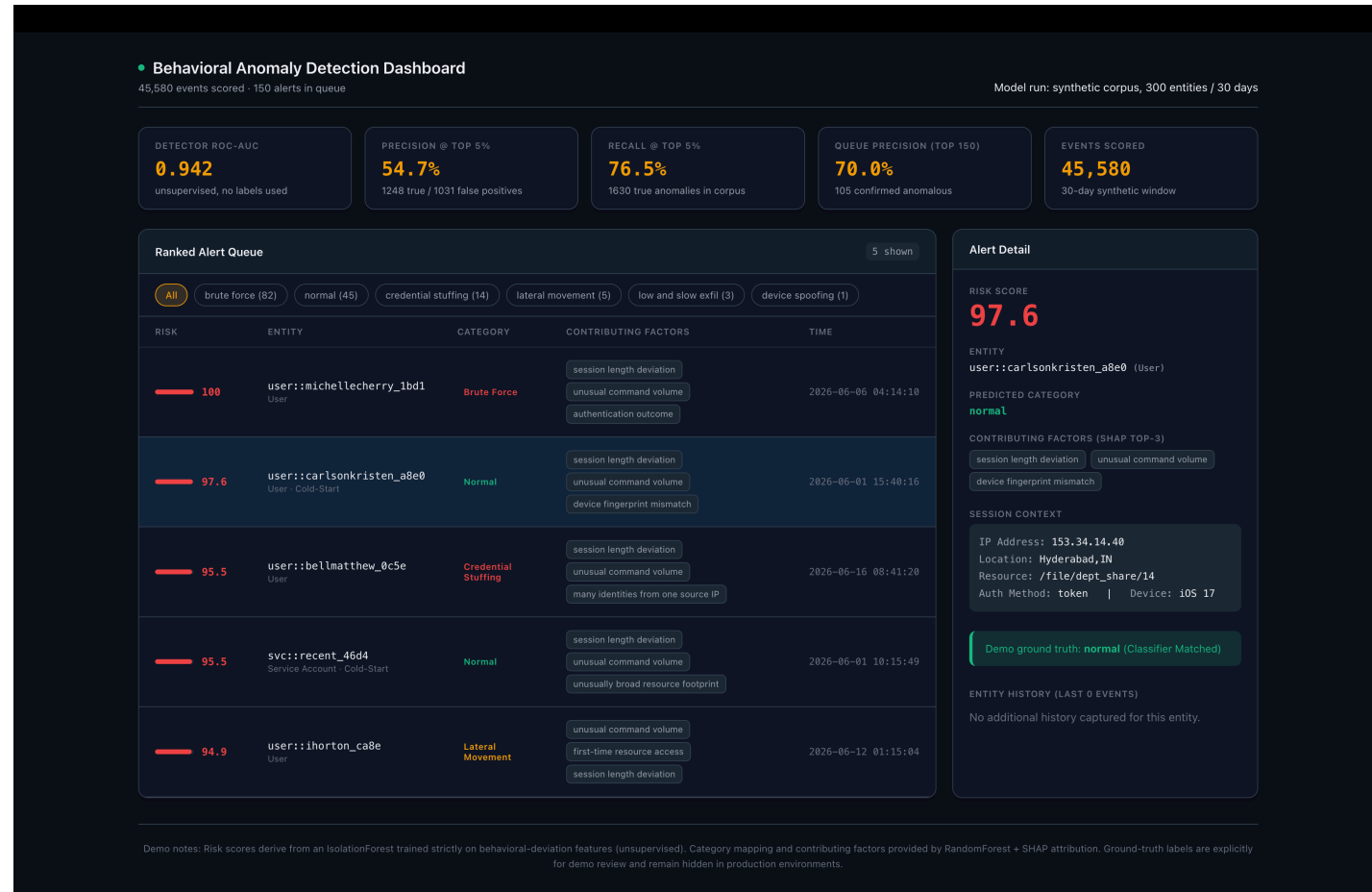
Treat "insider drift" as a human-review trigger, and add an explicit drift-detection trigger (like windowed re-baselining).

Port the running-statistics logic to a live stream processor for true online scoring in production.

ARTIFACTS

Relevant artifacts:

- **Codebase Embedded:** Fully working, single-threaded pipeline (generate_data.py, features.py, etc.) for exact reproducibility (seed=42).
- **Dashboard Snaps:** The dashboard.html interface showing a ranked alert queue, risk scores, predicted categories, SHAP chips, and entity history timelines.
- **Solution Proposal Snaps:** Generated report.docx and presentation.pptx detailing the architecture, metrics, and honest limitations.
- **Data Artifacts:** access_logs.csv and evaluation JSONs proving the pipeline's real output data.



RESEARCH AND REFERENCES

Details / Links of the reference and research work:

- **Core ML Research:** "Isolation Forest" (Liu, Ting & Zhou, 2008) for unsupervised detection, and "Random Forests" (Breiman, 2001) for classification.
- **Explainability (XAI):** "A Unified Approach to Interpreting Model Predictions" (Lundberg & Lee, NeurIPS 2017) for SHAP integration.
- **Graph Techniques:** Adaptation of Katz back-off smoothing to successfully blend entity-specific and population-level graphs for cold-start resilience.
- **Project Documentation:** Self-authored README.md, report.docx, and presentation.pptx detailing all assumptions, methodologies, and metrics.